

## CALCASIEU PASS, LA, USA

WMO: 997337

Lat: 29.7641N

Lon: 93.3428W

Elev: 10

StdP: 14.69

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 32.2       | 36.4 | N/A                           | N/A | N/A  | N/A | N/A | N/A  | 27.1                  | 47.4 | 25.3 | 47.6 | 13.4                  | 30   | 0.518 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |     |      |     |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|-----|------|-----|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%  |      | 2%  |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB  | MCDB | WB  | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k) | (l)  | (m) | (n)  | (o)                  | (p)  |
| 8             | 6.7                    | 89.2            | N/A  | 87.8 | N/A  | 87.0 | N/A  | N/A                 | N/A  | N/A | N/A  | N/A | N/A  | 9.7                  | 220  |

## Dehumidification DP/MCDB and HR

| 0.4%       |            |            | 1%         |            |            | 2%         |            |            | 0.4%       |            | 1%         |            | 2%         |            | Extreme<br>Max WB |
|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|-------------------|
| DP         | HR         | MCDB       | DP         | HR         | MCDB       | DP         | HR         | MCDB       | Enth       | MCDB       | Enth       | MCDB       | Enth       | MCDB       |                   |
| (a)<br>N/A | (b)<br>N/A | (c)<br>N/A | (d)<br>N/A | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A | (j)<br>N/A | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A | (n)<br>N/A | (o)<br>N/A |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |  |  |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|--|--|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |  |  |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |  |  |
| 26.0              | 23.7 | 20.7 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
|                   |      |      | DB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
|                   |      |      | WB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 69.0   | 51.5  | 54.8  | 61.9  | 68.5  | 75.8  | 81.9  | 82.7  | 83.7  | 79.6  | 71.5  | 60.1  | 55.5 |
|                                                                                  | DBStd       | 12.87  | 8.21  | 8.34  | 7.33  | 5.97  | 4.34  | 2.61  | 2.28  | 2.53  | 3.21  | 6.55  | 7.67  | 7.71 |
|                                                                                  | HDD50       | 195    | 82    | 48    | 10    | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 17    | 37   |
|                                                                                  | HDD65       | 1392   | 419   | 290   | 142   | 36    | 4     | 0     | 0     | 0     | 0     | 22    | 177   | 302  |
|                                                                                  | CDD50       | 7145   | 129   | 184   | 378   | 556   | 800   | 958   | 1013  | 1046  | 888   | 667   | 319   | 207  |
|                                                                                  | CDD65       | 2865   | 1     | 5     | 45    | 141   | 338   | 508   | 548   | 581   | 438   | 223   | 29    | 8    |
|                                                                                  | CDH74       | 29235  | 0     | 1     | 13    | 261   | 2375  | 6067  | 6877  | 7588  | 4498  | 1550  | 5     | 0    |
|                                                                                  | CDH80       | 9720   | 0     | 0     | 0     | 4     | 145   | 2072  | 2684  | 3353  | 1271  | 191   | 0     | 0    |
| Wind                                                                             | WSAvg       | 11.3   | 11.0  | 11.2  | 11.3  | 13.0  | 13.1  | 11.7  | 10.6  | 10.1  | 9.7   | 10.8  | 11.2  | 11.3 |
| Precipitation                                                                    | PrecAvg     | 61.2   | 5.9   | 3.8   | 4.0   | 4.6   | 4.1   | 6.7   | 6.1   | 4.7   | 5.2   | 5.2   | 4.8   | 4.8  |
|                                                                                  | PrecMax     | 83.5   | 13.5  | 8.6   | 10.1  | 9.1   | 12.7  | 10.0  | 9.6   | 10.0  | 12.9  | 20.8  | 14.0  | 9.9  |
|                                                                                  | PrecMin     | 41.8   | 1.4   | 0.7   | 1.8   | 0.5   | 0.0   | 1.4   | 3.0   | 0.5   | 0.7   | 1.5   | 0.6   | 2.4  |
|                                                                                  | PrecStd     | 11.6   | 3.4   | 2.5   | 2.2   | 2.4   | 3.2   | 2.4   | 1.9   | 2.6   | 3.1   | 4.7   | 3.1   | 2.2  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 67.5  | 72.2  | 75.5  | 79.4  | 84.8  | 91.1  | 89.4  | 93.6  | 89.8  | 85.4  | 75.4  | 71.4 |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 2%          | DB     | 65.5  | 69.5  | 74.0  | 77.4  | 82.6  | 87.8  | 87.4  | 89.7  | 87.2  | 83.5  | 73.4  | 69.4 |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 5%          | DB     | 64.3  | 66.4  | 72.7  | 76.5  | 81.5  | 86.4  | 86.6  | 88.4  | 85.9  | 82.2  | 71.9  | 68.1 |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 10%         | DB     | 62.7  | 64.9  | 71.1  | 75.5  | 80.6  | 85.6  | 86.2  | 87.4  | 85.0  | 80.8  | 70.4  | 66.3 |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 10.6  | 9.4   | 8.8   | 8.6   | 6.4   | 6.0   | 6.3   | 6.7   | 8.7   | 11.0  | 11.2  | 10.7 |
|                                                                                  |             | MCDBR  | 9.1   | 8.1   | 7.9   | 7.8   | 6.8   | 7.7   | 6.6   | 8.1   | 9.5   | 9.4   | 9.5   | 9.7  |
|                                                                                  |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 5% WB       | MCDBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.347  | 0.362 | 0.379 | 0.432 | 0.471 | 0.484 | 0.494 | 0.479 | 0.455 | 0.386 | 0.371 | 0.355 |      |
|                                                                                  | taud        | 2.518  | 2.487 | 2.430 | 2.279 | 2.204 | 2.217 | 2.221 | 2.267 | 2.298 | 2.476 | 2.456 | 2.495 |      |
|                                                                                  | Ebn at Noon | 281    | 287   | 288   | 275   | 263   | 258   | 255   | 258   | 261   | 275   | 270   | 272   |      |
|                                                                                  | Edh at Noon | 28     | 32    | 36    | 43    | 46    | 46    | 45    | 43    | 40    | 32    | 30    | 28    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 922    | 1063  | 1474  | 1783  | 2002  | 2017  | 1906  | 1797  | 1602  | 1425  | 1074  | 824   |      |
|                                                                                  | RadStd      | 94     | 171   | 137   | 137   | 128   | 147   | 150   | 152   | 168   | 171   | 102   | 90    |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (28 neighbors) | +0.59 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +193  | +148  |

Nomenclature: See separate page